MIL-C-48808(PA)
9 June 1975
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

CASTINGS, SAND, ALUMINUM BASE AND MAGNESIUM BASE (FOR SPECIAL WEAPONS APPLICATION)

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and the preparation for delivery criteria for three classes of sand castings entitled Castings, Sand, Aluminum Base and Magnesium Base (for Special Weapons Application) which are used with the Nike Hercules warhead section.

1.2 Classification. This specification covers castings of the following grades.

Grade A - Castings in which a defect will cause hazardous or unsafe conditions for individuals using or maintaining the product or cause a major end item not to perform its tactical function.

Grade B - Castings in which a defect may probably cause a hazardous or unsafe condition for individuals using or maintaining the product or may cause a major end item not to perform its tactical function.

Grade C - Castings in which a defect could result in failure or materially reduce the usability of a unit product for its intended purpose.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

Federal Test Methods Standards No. 151
Federal Standard Method QQ-M-151

MILITARY

MIL-I-6866	- Inspection, Penetrant Method of
MIL-I-8474	- Inspection of Aluminum Alloy Parts, Anodizing Process for
MIL-R-11471	- Radiographic Inspection of Metals.
MIL-C-25343	- Certificate of Penetrant Inspection Personnel

FSC NUOR

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Commercial

American Society for Testing Material

ASTM-B26-56T	- Specifications for Aluminum Base Sand Castings
ASTM-B80-56T	- Specification for Magnesium Sand Castings
ASTM-98-53T	- Radiographic Standards

STANDARDS

MILITARY

MIL-STD-1167	- Ammunition Data Card.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert
	- Ammunition Components.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.3 Surface conditions.

3.3.1 Visual. As cast or machined surfaces shall be free from cracks and cold shuts.

3.3.2 Contour. Areas of general depression shall be within the flatness or dimensional requirements specified on the detail drawings.

3.3.3 Machined surfaces.

3.3.3.1 Grade A and B castings. Unless otherwise specified on the detail drawings, machined surfaces of Grade A and B castings shall not contain porosity, holes and segregation exceeding those permitted by Tables II and III.

3.3.3.2 Grade C Castings. Unless otherwise specified on the detailed drawing, machined surfaces of Grade C castings shall not contain porosity, holes and inclusions exceeding that listed in Table I.

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TABLE I

Thickness of section in inches	Max average dimension of defects, percent of section thickness	Largest individual defect, percent of section thickness
3/8	10	20
over 3/8 to and including 1	7	14
over 1	5	10

3.4 Internal conditions.

3.4.1 Radiographic examination.

3.4.1.1 Segregated discontinuities. The interior of the castings shall show no evidence of cracks or linear shrinkage, or show evidence of gas holes, shrinkage cavities, porosity, foreign materials, or segregation exceeding those in Tables II and III. Unless otherwise specified on the drawings, ASTM E98-53T radiographic standards are applicable. Grade A castings shall be evaluated as oritical all over unless noted on the drawing to the contrary. Grade B castings shall be evaluated as non-oritical all over unless noted on the drawing to the contrary.

TABLE II

Aluminum Castings

Defect	Section Thickness	Section No	Radiographic No Critical	Radiographic No Non Critical
Gas holes	All	GA - 1.1	2	4
Gas porosity, round	over 3/4 inch	GA - 1.2	3	6
	3/4 inch and under		3	4
Gas porosity, elongated	All	GA - 1.22	3	5
Shrinkage cavities	All	SA - 2.1	3	5
Shrinkage sponge	3/4 inch and under	SA - 2.2	3	5
	over 3/4 inch	SA - 2.2	4	6
Foreign materials	3/4 inch and under	HA - 3.11	3	5
	over 3/4 inch	HA - 3.11	5	7
Segregation	All	HA - 3.2	1	1

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TABLE III

Magnesium Castings

Defect	Section Thickness	Section No	Radiographic No Critical	Non Critical
Gas holes	All	GM - 1.1	2	3
Gas potosity	All	GM - 1.2	1	1
Shrinkage cavities	All	SM - 2.1	2	2
Microshrinkage (all types)	All	SM - 2.3	4	5
Foreign Materials	All	HM - 3.11	4	5

3.5 Acceptance criteria. The predominating defect shall govern in deciding the acceptability of the area during radiographic inspection.

3.6 Marking. Castings shall be identified with the pattern producing them. Castings shall also be identified with respect to melt and/or heat treatment. Additional marking shall be as specified on the detail drawings (see 6.1). Tensile test specimens shall be marked with melt number and heat treat lot number as applicable. In the event marking is impractical, the contracting officer shall be so notified and written authorization for deletion of this requirement requested.

3.7 Repair of castings. Castings shall not be repaired by welding, impregnation, peening, blending, or any other method without permission of the procuring agency. Such permission may be given only when the repaired defect will not adversely affect the strength or machinability or otherwise impair the suitability of the castings for the purpose intented. Under no conditions, shall soldering be permitted for repairing castings. Repaired castings shall be reinspected in accordance with the applicable portions of the pertinent specifications, drawings and directives.

3.8 Workmanship.

3.8.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.8.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.8.3 Burrs. Parts shall be free from burrs.

3.8.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.8.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 castings or one casting from each cavity pattern used, whichever is greater, for first article testing to the testing gacility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table IV. Tests shall be conducted in order specified in Table IV.

Table IV

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Visual (see 3.3.1)		X
2. Surface conditions (see 3.3.3)		X
3. Radiographic inspection (see 3.4.1)		X
4. Tensile test specimens see	X	
5. Mechanical property	X	
6. Heat treatment test		X
7. Chemical Analysis		X

4.2.3 Frequency. The preproduction sample shall be submitted, prior to the start of regular production, upon introduction of new patterns, or molds, upon changes in gating or riser practice, when any other change in process occurs which potentially changes the casting, or when specified by the procurement agency.

4.2.4 Representation of sample. The preproduction sample shall be representative of the techniques, process, equipment and controls to be used for castings for delivery to the procurement agency.

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4.2.5 Rejection. If any item failt to comply with any of the applicable requirements, the first article shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation. See 6.4.1.

4.3.2 Testing. Each casting shall be subjected to all of the inspection and tests specified herein and on the applicable drawings, details specifications and contracts. Tests shall be performed in the order specified below where applicable:

Test	Test classification
a. Visual	Major
b. Surface inspection	Major
c. Radiographic inspection	Major
d. Tensile test	Major
e. Mechanical properties	Major
f. Heat treatment test	Major
g. Chemical analysis	Major

Of these tests d, e, f and g are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3 Sampling.

4.3.3.1 Method. All samples shall be selected by the Government inspector.

4.3.3.2 Sample size for radiographic inspection.

4.3.3.2.1 Grade A castings. Radiographic inspection for Grade A castings shall be conducted 100 percent.

4.3.3.2.2 Grade B castings. Sample casrings shall be selected in accordance with Table V.

TABLE V

Lot size	Sample size
1	1
2	2
3-4	3
5-7	4
8-11	5
12-17	6
18-27	7
28-48	8
49 and over	9

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4.3.3.2.2.1 Rejection and retest. If any casting fails to comply with the applicable requirements the lot shall be rejected subject to retest. The contractor shall perform 100 percent radiographic screening on the rejected lot at his own expense and disoretion. Acceptability of each casting of the resubmitted lot shall be based on the contractors radiographs. Items failing to meet requirements during the retest shall be rejected. A second retest shall not be permitted.

4.3.3.3 Sample size for cast tensile specimens.

4.3.3.3.1 Sampling. Specimens from each lot shall be selected and action taken in accordance with specification Federal Test Method Standards - 151.

4.3.3.4 Sampling size for heat treatment of tensile test specimens.

4.3.3.4.1 Sampling. The quantity of specimens to be heat treated with their respective lot of casting, shall be sufficient to assure the availability of test specimens for one reheat treatment if so desired.

4.3.3.4.2 Acceptance. Acceptance shall be based on the first acceptable specimen, in accordance with Federal Test Method Standards - 151, complying with the applicable drawing requirements.

4.3.3.4.3 Retest. Any heat treat lot that fails paragraph 4.3.3.4.2 may be reheat treated once with the remaining test specimens outlined in paragraph 4.3.3.4.1 and retested for acceptance. No new test coupons may be introduced in the reheat-treat process. A second retest will not be permitted.

4.3.3.5 Sample size for chemical analysis.

4.3.3.5.1 Sampling. Samples shall be selected and action taken in accordance with Federal Test Method Standards - 151.

4.4 Test procedures.

4.4.1 Visual. Each casting in the lot shall be inspected for compliance with paragraph 3.3.1 and with dimensional and surface finish requirements.

4.4.2 Surface Inspection. Castings inspected for compliance with paragraph 3.3.3 shall be tested as specified in Specification MIL-I-6866. Fluorescent or non-fluorescent (dye) penetrant may be used. Anodizing inspection as specified in MIL-I-8474 may be substituted.

4.4.3 Radiographic inspection. The representative castings shall be tested as specified in MIL-R-11471.

4.4.4 Tensile test specimens.

4.4.4.1 Two or more test bars shall be cast from a melt in the lot, to represent that lot. No cast test bars are required if conformance to mechanical property requirements is demonstrated by test of specimens cut form castings in accordance with 4.4.11.

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4.4.4.2 The test bars from each lot shall be heat treated to the same temper simultaneously with castings of the same lot. When test bars are used, one of the bars, representative of the particular lot, shall be tested and the lot judged by the results of this test.

4.4.4.3 Each heat treating furnace charge shall be represented by at least one set of the test bars. The heat treatment of a furnace charge shall be judged by these test bars.

4.4.4.4 When test specimens from castings are required in accordance with 4.4.11 one or more castings in the temper to be furnished shall be selected from each lot and not less than three tension test specimens shall be machined from each casting, one of which shall be from the thinnest section, one from the thickest and one from an intermediate section.

4.4.4.5 Specimens cut from castings shall be in the form of flat or round tensile test specimens conforming to the requirements for specimen type 1, 2, 4 or 5 of Federal Standard Methods - 151 or any other types of specimens designated on the drawings, contract or purchase order, or acceptable to the inspector.

4.4.4.6 Unless otherwise specified, test bars shall be cast in accordance with the form and dimensions of either Figure 11A or 11 B of Federal Standard Methods QQ-M-151. If Figure 11A is used, a skimmer core may be provided in the mold. The contractor's own test bar design may be used provided it is acceptable to the procuring agency. Tolerances shall be within limits of good casting practice.

4.4.4.7 In the event the metal for castings is given any treatment such as fluxing, or cooling and reaheating, the metal for the test bars shall form a portion of the metal so treated. The temperature of the metal while pouring the test bars shall not be lower than 20 degrees Fahrenheit (°F) below the temperature at which the castings are poured. Test bars shall not be lower than 20°F below the temperature at which the castings are poured. Test bars shall not be chill-cast.

4.4.4.8 When the design of castings is such as to require pouring temperatures in excess of 1400°F, for the production of a sound product, and conformance with 4.4.4.7 is not practicable, the conditions of test bar pouring and/or mechanical properties to be met by the test bars shall be as specifically approved by the procuring agency.

4.4.4.9 The inspector shall, at his discretion, witness the pouring of the test bars and their removal from the sand, and shall stamp each test bar for future identification.

4.4.4.10 Unless otherwise specified, test bars shall be tested without machining except as it may be necessary to adapt them to the grips of the testing machine.

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4.4.4.11 Mechanical property requirements shall be determined by tests on specimens cut from castings when:

a. Specified on the drawings
b. Authorized by procuring agency
c. Required by the inspector in event of inferior quality in castings of a particular lot for which representative separately cast test bars have met the mechanical property requirements of this specification.

4.4.4.12 Tensile requirements. The required mechanical properties may be specified by the contracting officer, for each particular alloy designation. When no other particular specification of tensile properties are required the mechanical properties shall conform to those specified in ASTM-B-26-56T or B80-56T as applicable. Tension tests shall be performed in accordance with Federal Standard Methods 151. The yield strength shall be determined by the offset method or the extention-under-load method. If the offset method is used, the limiting set shall be 0.2% (.022 inch per inch of gage length); if the extension-under-load method is used, the limiting extension shall be 0.0075 inch.

4.4.5 Heat treatment test. The representative specimens shall be so located so they receive approximately the same quenching rate as do the castings they represent.

4.4.6 Chemical analysis. Chemical analysis shall be made by wet chemical or spectrographic methods. In case of dispute, chemical analysis by the wet chemical method shall be the basis for acceptance.

4.4.6.1 Chemical Constitution: The limiting or maximum amounts of constituent elements may be specified by the contracting officer for each particular alloy composition. When other particular specification of chemical analysis is not required, the composition shall conform to the ASTM-B26-56T, Tentative Specifications for Aluminum Base Alloy Sand Castings or ASTM-B80-56T, Specification for magnesium sand castings as applicable.

4.4.6.2 Chemical analysis of the individual melts, by the contractor, may be waived by the inspection, provided that the foundry method of composition control is acceptable to him, or that all the material in the lot can be identified as being from melts previously analysed and found to be in conformance with the chemical requirements of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The casting shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

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5.1.2 Level C. The casting shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.2 Packing.

5.2.1 Levels A and C. The casting shall be packed in accordance with the applicable instructions contained in MIL-STD-1169.

5.3 Marking. Marking shall be in accordance with MIL-STD-1169.

5.4 Data cards. Data cards shall be prepared in accordance with MIL-STD-1167.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Grade required (see 1.2).
c. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).
d. The location of locations which are not acceptable for raised or impression marking when the improper location of such marking could have a deleterious effect on the static or fatigue strength of the part or is important because of subsequent machining operations. (Impression markings improperly located may cause failures).

6.1.2 Contract data requirements (see 5.4). Data specified in 5.4 will be delivered as identified on a numbered DD Form 1664 when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Definitions.

6.3.1 Lot. A lot shall consist of not more than 1000 pounds of cleaned castings and tensile specimens of the same composition and temper, same drawing, same drawing revision, same specification, same specification revision submitted for inspection at one time. Unless otherwise specified, each lot of castings shall be from the same melt or group of melts as may be designated by the inspector.

6.3.2 Melt. A melt is an unspecified quantity of metal melted in any single foundry furnace at one time and poured into molds in normally rapid succession.

6.4 Supersession data. This specification supersedes the requirements of Purchase Description X-PA-PD-994 dated 11 June 1957.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. NUOR-A036

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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